

Problem M. Hardcore String Counting

Input file: *standard input*
Output file: *standard output*
Time limit: 15 seconds
Memory limit: 512 mebibytes

You are given a non-empty string s of lowercase English letters. A string w of lowercase English letters is *good* if every proper prefix of w does not contain s as a substring, but w itself does.

Find the number of good strings of length m . Because this number can be very large, output it modulo prime number $998\,244\,353 = 2^{23} \cdot 119 + 1$.

Input

The first line of the input contains two integers: n , the length of s , and m , the length of strings you have to count ($1 \leq n \leq 10^5$, $n \leq m \leq 10^9$). The second line contains a string s consisting of n lowercase English letters.

Output

Output a single nonnegative integer: the number of good strings of length m modulo $998\,244\,353$.

Examples

<i>standard input</i>	<i>standard output</i>
6 7 aaaaaa	25
3 5 aba	675